Simulation and Power Spectral Density Analysis of TLS-Induced Noise in Superconducting Resonators

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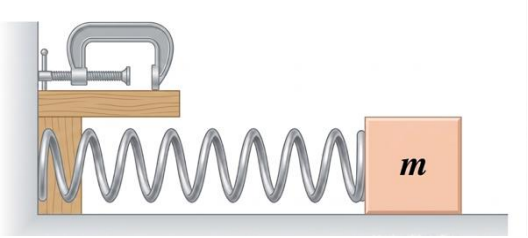
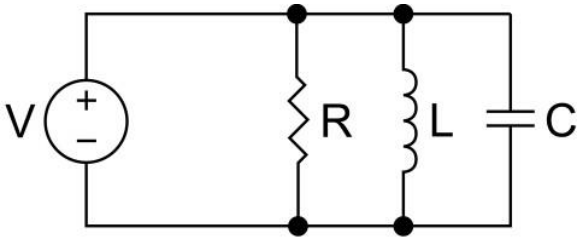
What is a Resonator?

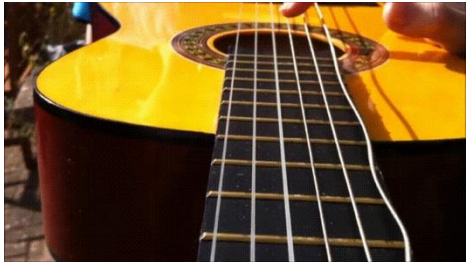
- A resonator is a system that stores energy and oscillates at a specific frequency
- In normal (lossy) systems:
  - Energy dissipates (resistive heating, damping)
  - Oscillations decay quickly
  - Broad resonance peak (low Q)

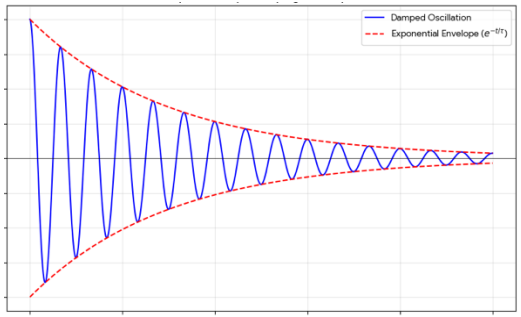




$$\ddot{q} + R\dot{q} + \frac{1}{C}q = 0$$

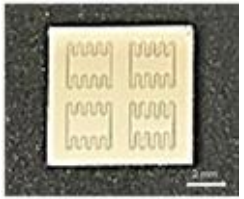
$$\rightarrow q(t) = q_0 e^{\frac{-R}{2L}t}\cos(\omega t)$$

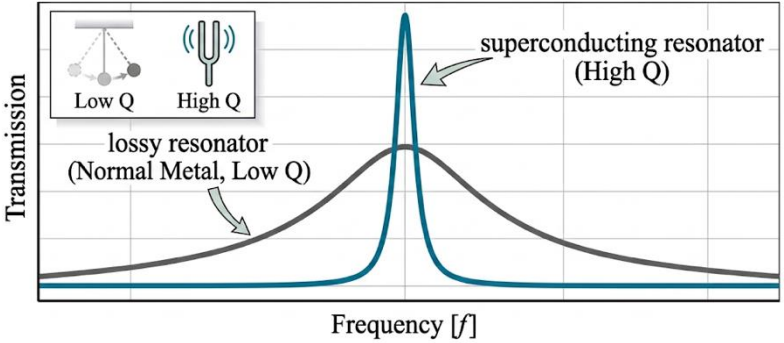




Superconducting Resonators

- ❑ Same resonator concept (LC circuit), but made from a superconductor
- ❑ Key difference: R ≈ 0 → minimal energy loss





$$Q = \frac{\omega_0 L}{R} \rightarrow \infty \text{ as } R \rightarrow 0$$

$$\text{Linewidth } \Delta\omega = \frac{\omega_0}{Q} \text{ very small}$$

- Very low loss → long-lived oscillations
- Extremely sharp resonance
- Tiny frequency shifts become measurable

- ❑ Applications:
  - ❑ Sensitive detectors (photons, temperature, materials)
  - ❑ Quantum circuits (store & read out quantum states)

→ Performance limited not only by loss, but also by noise

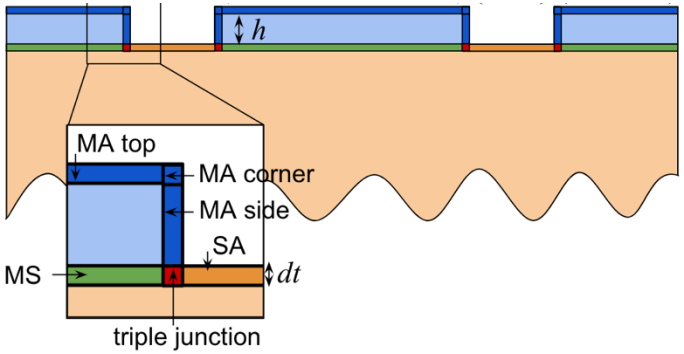
Noise Mechanisms in Superconducting Resonators

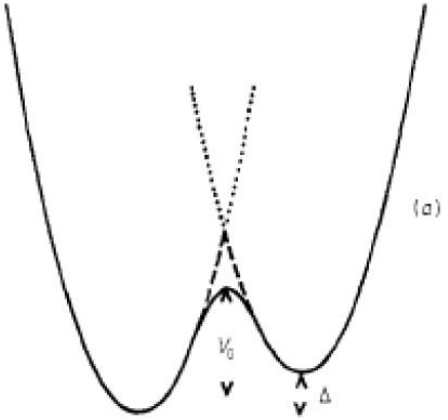
- Thermal noise
  - from $k_B T$
- Quasiparticles
  - broken Cooper pairs
- Readout noise
  - from electronics
- Two-level System (TLS)
  - microscopic defects
  - weakly suppressed at low T
  - frequency noise + dissipation

- At millikelvin temperatures:
  - Thermal noise → strong suppressed $n_{th} \sim e^{-\hbar\omega/k_B T}$
  - Quasiparticles → exponentially suppressed $\sim e^{-\Delta/k_B T}$
  - Readout noise → not intrinsic (can be calibrated or reduced)
  - TLS → remain active and dominate

TLS don't freeze out, and we can't easily get rid of them

Two-level System (TLS)





- What is TLS?
  - Defect with two stable configurations
  - Modeled as a double-well potential
- Dynamics
  - Quantum tunneling + environment interactions
  - Random switching: $s_i(t) = \pm 1$
  - → random telegraph signal
- Coupling to resonator
  - Electric dipole → interacts with resonator field
  - Changes dielectric environment
- Effects
  - Frequency fluctuations (noise)
  - Dissipation (loss)

Project Objectives

- ❑ Model noise using Monte Carlo simulation
  - ❑ Simulate stochastic switching of a single TLS
  - ❑ Combine many TLS contributions
- ❑ Add white noise (thermal/readout)

→

- ❑ Generate time-domain signal $\delta f(t)$
- ❑ Mimics experimental frequency fluctuations

$$\delta f(t) = \sum_i s_i(t)\,\Delta f_i + \textit{white noise}$$

$$s_i(t) = \pm 1$$

- ❑ Analyze using PSD
- ❑ Connect microscopic TLS dynamics → observable noise

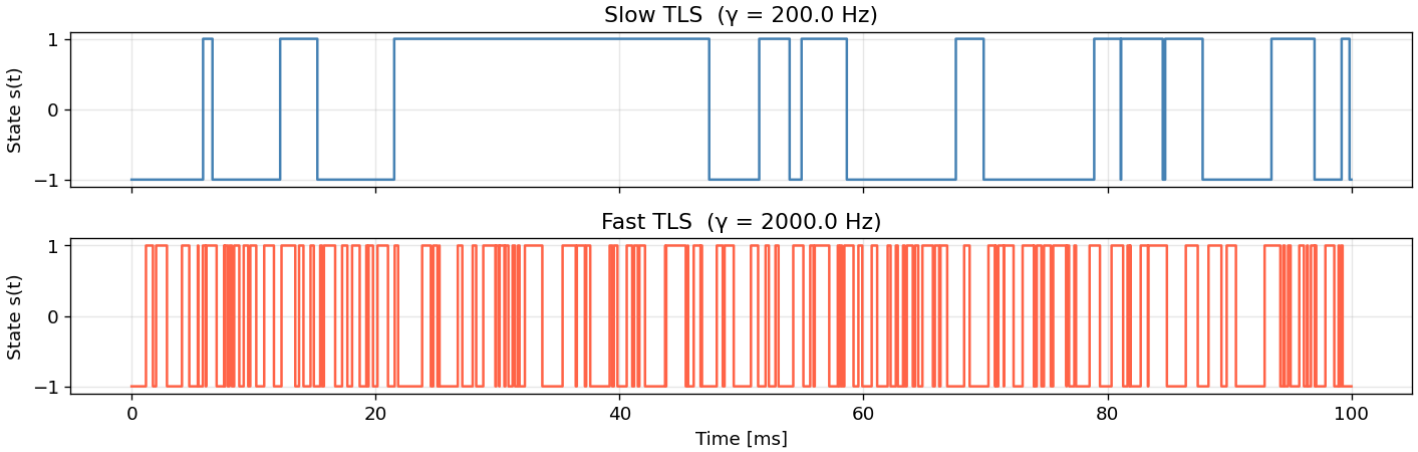
We simulate microscopic TLS switching to reproduce and analyze macroscopic noise

Noise Simulation: Simulating a Single TLS

- Set up simulation grid
  - Total time T (capture slow noise)
  - Time step $\Delta\mathrm{t}$ (resolve fast switching)
- TLS model
  - Two states: $\mathrm{s(t)}=\pm 1$
  - Frequency shift: $\pm\Delta\mathrm{f}$
- Stochastic switching
  - Flip probability: $p = \gamma \Delta t$

<u>(Simplified) Algorithm</u>

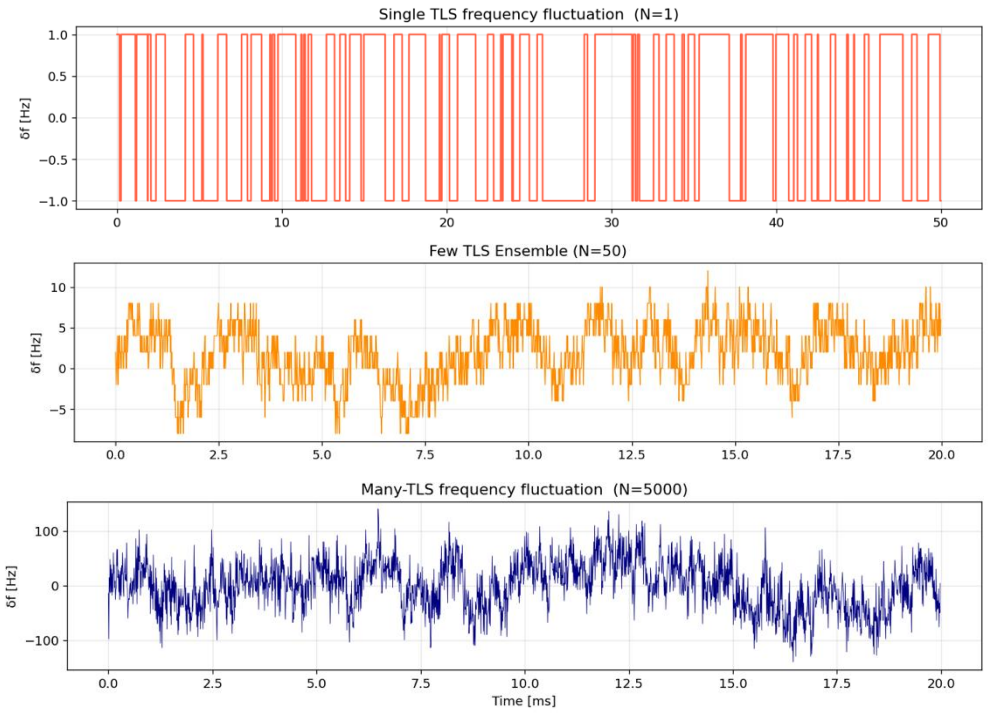
- At each time step:
  - Draw random number $r \in [0, 1]$
  - If $r < p \rightarrow$ flip state
  - Else $\rightarrow$ stay the same



Noise Simulation: Simulate Ensemble of TLSs

- Simulate N TLS independently
- For each TLS, assign:
  - (1) coupling strength Δf_i
  - (2) switching rate γ_i
- Run the same Monte Carlo update for each TLS
- Sum contributions:

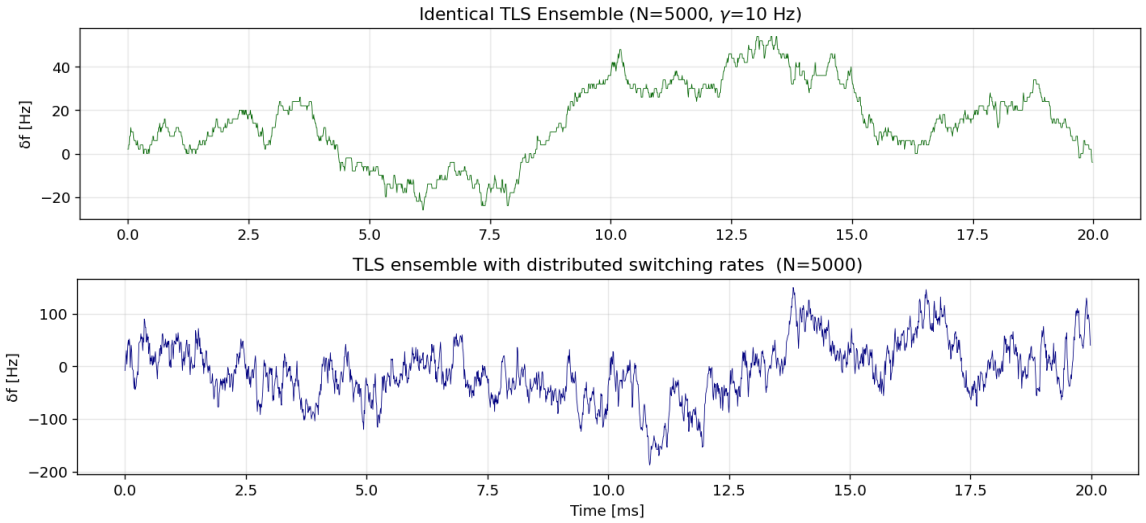
$$\boxed{\delta f(t) = \sum_{i=1}^{N_{\mathrm{TLS}}} s_i(t)\, \Delta f_i}$$



Superposition of many random TLS → complex noise signal

Noise Simulation: Effect of TLS Parameter Distributions

- Case 1: Identical TLS
  - Same Δf and same γ
  - All TLS behave the same
  - → coherent Lorentzian noise
- Case 2: Distributed switching rates (STM)
  - $\gamma_i \epsilon\ [\gamma_{\min}, \gamma_{\max}]$ (log-uniform distribution)
  - Same Δf_i
  - →broad range of timescales
- Coupling strength:
  - kept constant for clarity
  - can be broadly distributed in real materials

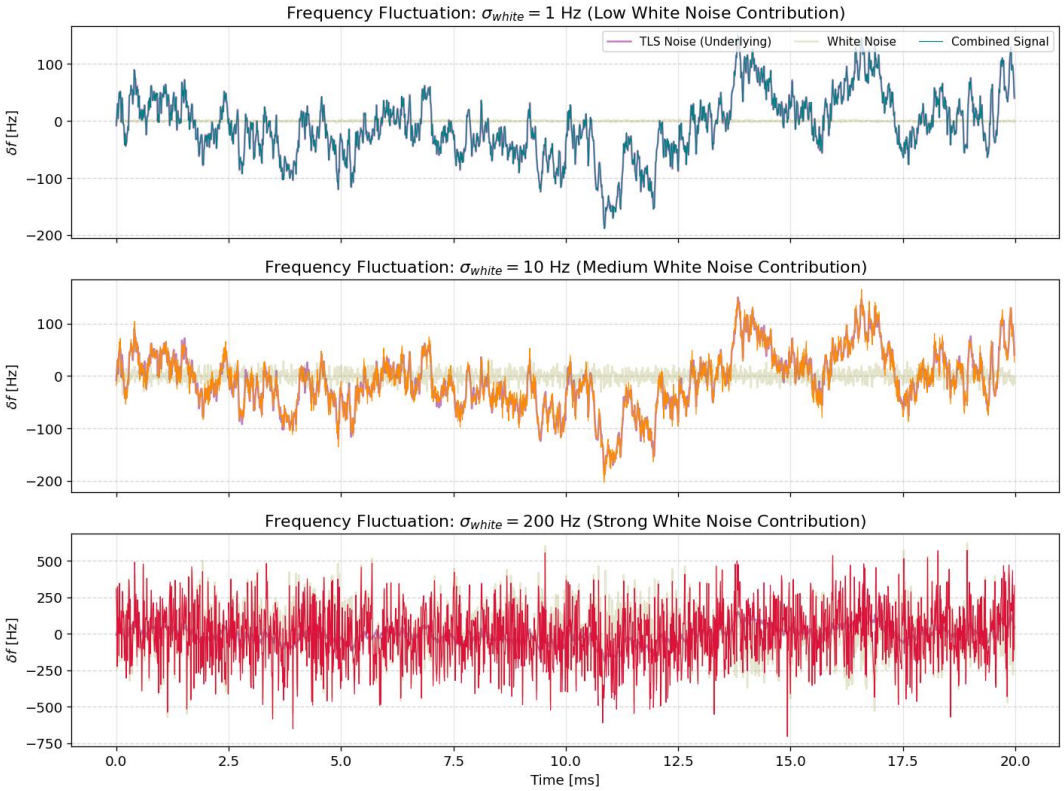


- Identical TLS → single-timescale random telegraph signal (structured fluctuations)
- Distributed γ_i → irregular, multi-timescale fluctuations

Noise Simulation: Adding White Noise

- Add random fluctuations at each timestep
- Gaussian noise: $\mathcal{N}(0, \sigma^2)$
  - Models thermal / readout noise
  - Zero mean → no bias
  - σ sets noise strength

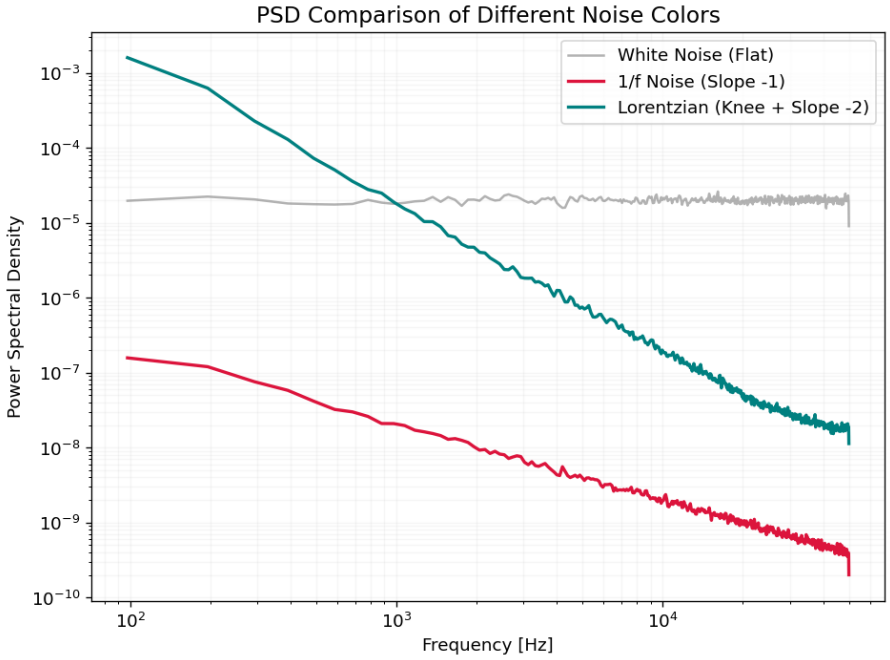
$$\delta f(t) = \delta f(t) + \mathcal{N}(0, \sigma^2)$$



White noise sets the high-frequency background floor of the signal

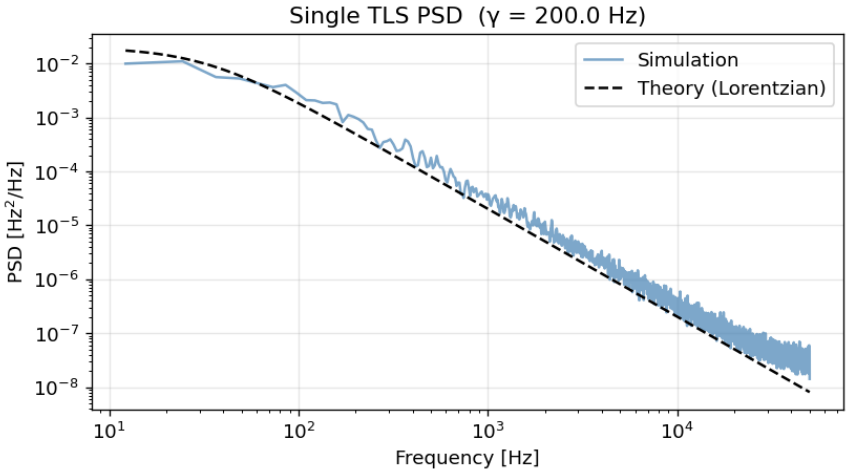
Noise Analysis: Power Spectral Density (PSD)

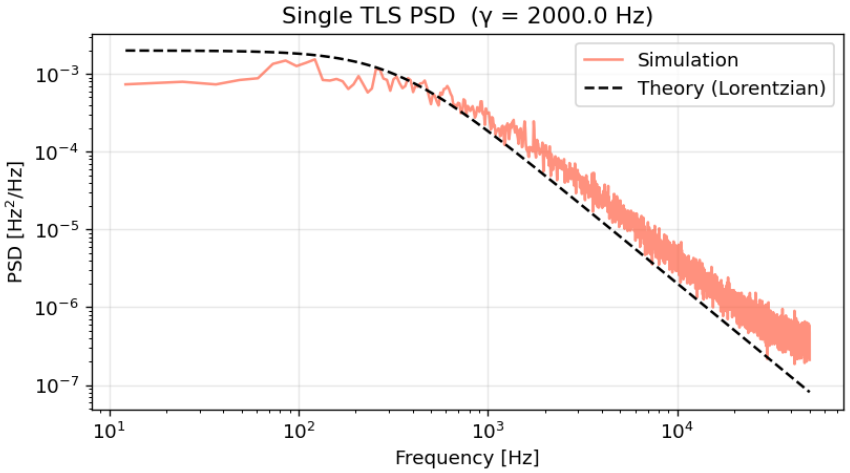
- PSD = how signal power is distributed across frequencies
- Intuitively, it tells us which time scales dominate the fluctuations:
  - Low frequency → slow fluctuations / drift
  - High frequency → fast fluctuations / noise
- Used to identify noise mechanisms in superconducting resonators



Noise Analysis: Single TLS

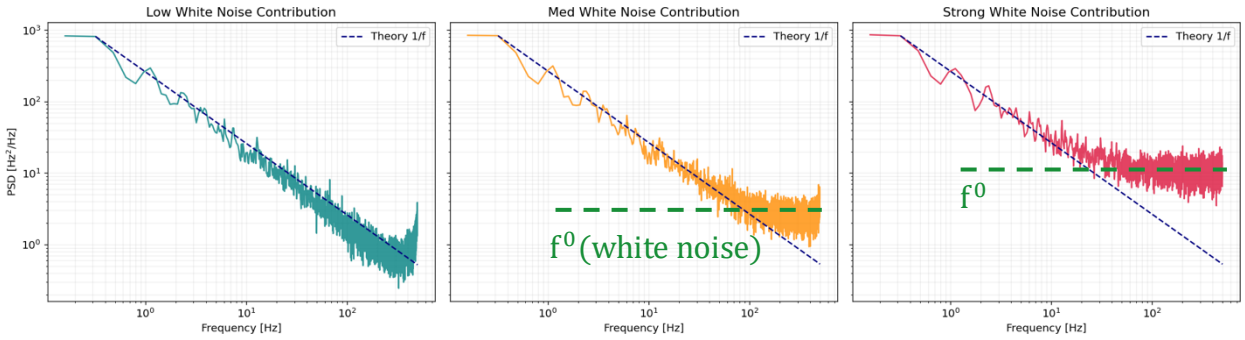
- PSD is Lorentzian:
  - flat at low f
  - roll-off at $f \sim \gamma$
- Slow TLS → Lorentzian shifts left
- Fast TLS → Lorentzian shifts right



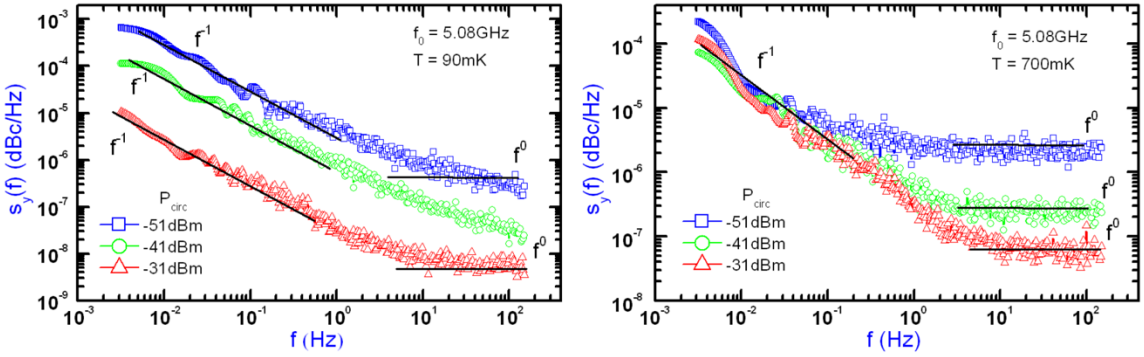


Noise Analysis: Effect of White Noise on PSD

- Low white noise → PSD follows $1/f$ over wide range
- Increasing white noise → high-frequency flattening
- Transition frequency set by relative noise strengths



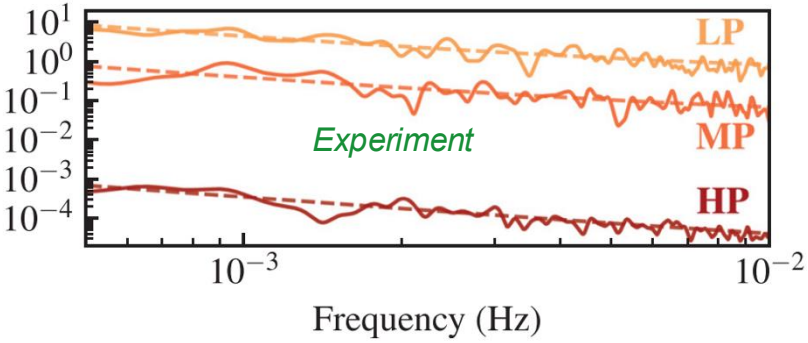
Numerical

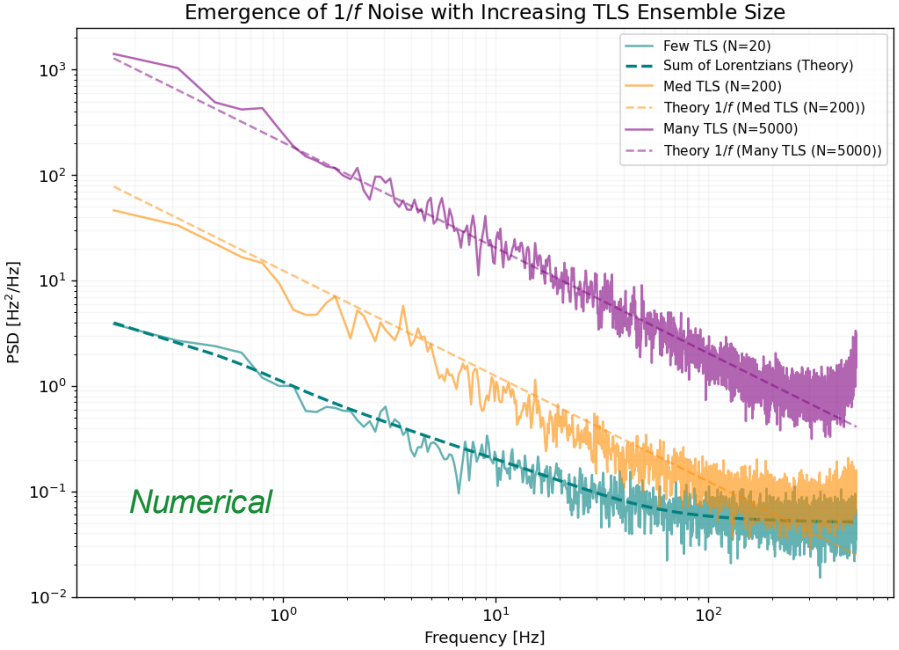


Experiment: Burnett et al. (2018)

Noise Analysis: Different Size of TLS Population

- Many TLS → overlapping Lorentzians → emergent $1/f$
- Increasing TLS density → stronger noise floor





- TLS noise decreases with increasing drive power
- High power → TLS saturation

Summary and Future Work

- What we did
  - Simulated frequency noise from independent TLS + white noise
  - Generated time-domain signals using stochastic switching
  - Computed PSD to identify noise signatures
- What we showed
  - Single TLS → Lorentzian PSD
  - Many TLS → emergent $1/f$ noise
  - White noise contribution modifies high-frequency behavior
  - Increasing drive power → TLS saturation → reduced noise (experimentally consistent)

<u>Extension</u>

- Beyond STM (GTM)
  - Include interactions between TLS
  - Expect deviations from pure $1/f$, non-Gaussian noise
- Additional analysis
  - Allan deviation (ADEV) → time-domain stability